FATIRO AR

ARCHITECTURAL SIMULATION - KIGALI, RWANDA

Technical documentation: AR simulation technical implementation

Summative Assignment 2 · builds on the Assignment 1 SRS and Data Management Plan

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PLATFORM	Android 10+ (API 29), ARCore, ARM64	ENGINE	Unity 6000.4.7f1, AR Foundation 6.3, URP 17
REPOSITORY	GitHub link	BUILD	APK link FatiroAR.apk, 57 MB
VIDEO	Project Walkthrough Video / Code Video	TESTS	39 EditMode, 2 PlayMode, all passing

01 WHAT THE APP DOES

Fatiro (from urufatiro, Kinyarwanda for "foundation") puts a sloped Rwandan hillside on a real table through the phone camera. The learner picks a house, a slope (0 to 30°) and a soil, anchors the plot on a detected plane, and builds the house as a hologram, one building component at a time. Opening the test bench reveals the real house: a scan line rises through it and every piece turns into brick, concrete, glass or roof sheeting. The learner then shakes, blows, rains on or crowds the house. Members past their capacity turn red and fall, bringing down whatever rests on them. The learner reads the stress chart, reinforces the weak point and tests again; the session is scored and saved.

Flow: Start → Setup → Placement (plane detection, anchor) → Building (hologram) → Test bench (reveal) → Load test ⇄ Review → Score, save, history.

02 DESIGN DIRECTION AND REFERENCES

The houses and the interface come from the reference boards I collected on Pinterest. Two ideas shaped the build. First, in biomimetic architecture the animal or object **is** the building's skin: the falcon pavilion, the violin concert hall, the manta-ray habitat and the shell pavilion are smooth white shells with bands of glass over a real structure. Fatiro's panther, elephant and car houses follow that rule, a ferrocement shell over a concrete and brick home. Second, the violin sheet shows a design moving from object to wireframe, to structure, to finished skin. That sequence became the app's own process: wireframe hologram, component-by-component structure, then the reveal. The holographic-table image set the visual theme used in the app and in this document.



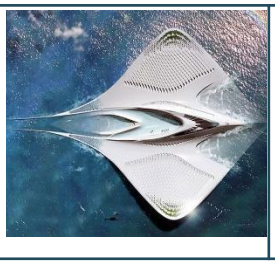
Holographic table: theme, world label, rings



Violin hall: object → frame → skin



Falcon pavilion: the animal is the roof



Manta habitat: shell with glazing bands



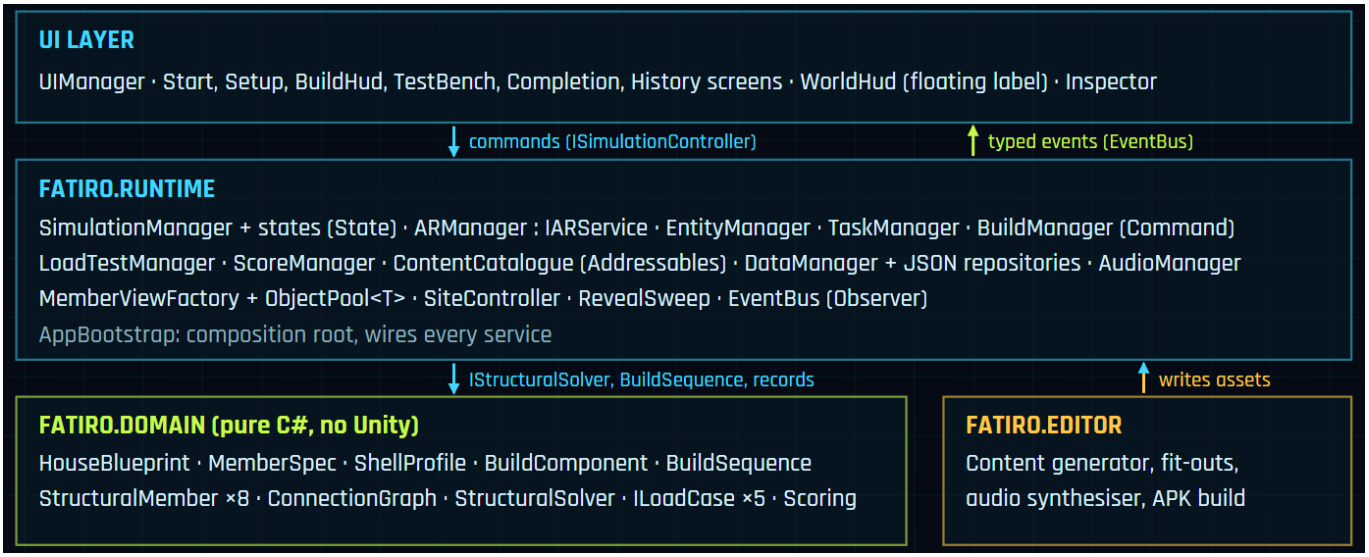
Shell pavilion: lofted ribs and caps

Six houses, six lessons

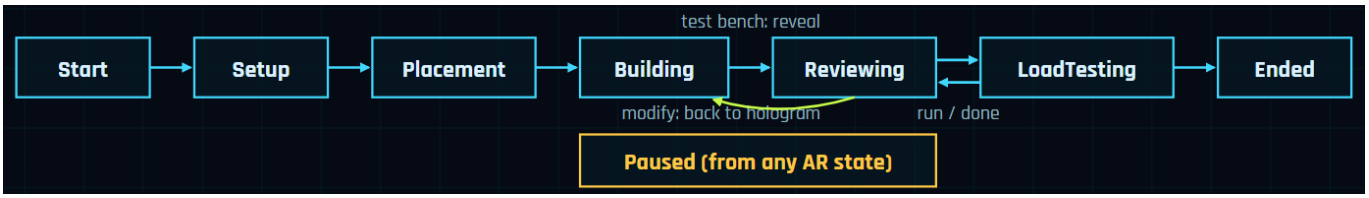
HOUSE	WHAT IT IS	FAILS WHEN	FIXED BY
Ingwe PN-01 panther	Black ferrocement shell on concrete rib arches; split-level floors; viewing room in the raised head	Quake topples unbraced ribs; people overload the head cantilever	Flank braces; foreleg props
Inzovu EL-01 elephant	Home on the belly floor under a dome, on four concrete legs; ear canopies; trunk downpipe	Wind lifts the ears; the trunk overflows and the roof floods	Ear tie-downs; steel trunk
Imodoka CR-01 car	A sports-car body on raked concrete stilts; kitchen behind the windscreen	Quake buckles the long downhill stilts	Downhill bracing
Urugo BG-01 bungalow	Concrete frame, fired-brick walls, iron-sheet roof	Wind lifts nailed iron sheets	Storm straps
Igorofa TS-01 two storeys	Concrete stair, clay-tile roof, balcony over the valley	A crowd overloads the balcony cantilevers	Balcony props
Amacumbi AP-01 flats	Three storeys of flats over open parking	Quake: the open floor is a soft storey	Shear walls

03 FINAL ARCHITECTURE

The code is split into three assemblies so the engineering model can be tested without Unity. Fatiro.Domain is pure C# with **noEngineReferences**: house blueprints, members, the solver, load cases, build-sequence rules, scoring and the session record. It compiles in a plain .NET test project, which is how the solver was calibrated. Fatiro.Runtime is the Unity layer: AR, views, UI, audio, persistence and Addressables. Fatiro.Editor generates every asset from code (materials, meshes, prefabs, synthesised audio, design assets, Addressables groups, the scene and Android settings), so the project rebuilds from the repository with one menu command. **AppBootstrap** is the composition root: it creates every service and passes dependencies in, with no singletons. Commands go down through interfaces; notifications come back up as events.



Simulation states



Each state is a class that accepts only the input that is legal at that moment, so a load test cannot run during placement and the structure cannot be edited during a test. Losing tracking for more than a few seconds moves to Paused, which remembers the previous state and resumes it.

04 OOP IMPLEMENTATION

- **Inheritance and polymorphism.** **StructuralMember** is abstract; **Foundation**, **Column**, **Beam**, **Arch**, **Cantilever**, **Brace**, **Panel** and **Drain** each override how they resist load (**VerticalCapacity**, **LateralCapacity**, **UpliftCapacity**, **LateralStiffness**). A column buckles with length, a cantilever weakens with the square of its reach, a roof panel is held down by its own weight plus nails or storm straps. The solver never asks what type a member is.
- **Encapsulation.** A member's state and utilisation change only through **ApplyResult**; the solver returns results and never mutates members, which a unit test checks. **MemberSpec**, **ShellProfile** and **BuildComponent** are immutable.
- **Composition.** A **HouseBlueprint** is composed of member specs, lofted shells and build components; a **ConnectionGraph** holds placed members and their load paths. Views (**MemberView**) only draw: they own no rules.
- **Data-driven entities.** Designs are **HouseDesign** ScriptableObjects (serialised specs, shells, components) loaded through Addressables, so a new house is new data, not new code (NFR-12).

05 SIMULATION RULES

Each member has a kind, a material and a geometry that depends on the site. On a slope, ground-anchored points drop by tan(slope) per metre, so downhill stilts get longer and more slender. A load test applies dead load and the chosen case: occupancy on floor areas, wind pressure on every surface and uplift on roofs, rainfall on roofs (ponding if the drains overflow), or ground shaking proportional to the weight above the foundations. Loads are pushed from the roof down through each member's supports; sideways force is shared by stiffness between the members that stand on footings and the braces or shear walls between them. Utilisation is demand divided by capacity. Members above 100% fail, members left without support fail in the next wave, and the solver repeats until nothing more fails, which gives a progressive collapse animated wave by wave.

The houses are built from five member shapes plus a lofted shell: an architect-style section swept through stations along the building (a superellipse arch from base to crown). Each shell patch, such as a flank, a band of glass or an end cap, is a real member with its own measured area, so the panther's skin carries wind and rain like any wall or roof. Classic houses use the same tool for their gable roofs.

DEFAULT TEST AT 20° ON LATERITE	BEFORE	AFTER THE FIX
Imodoka, earthquake level 3	**113%, fails**	**70%** with downhill bracing
Ingwe, occupancy 3 kN/m² in the head room	**115%, fails**	**30%** with foreleg props
Inzovu, wind 120 km/h	**182%, fails**	**57%** with ear tie-downs
Urugo, wind 120 km/h	**158%, fails**	**69%** with storm straps
Igorofa, occupancy 3 kN/m² on the balcony	**137%, fails**	**76%** with balcony props
Amacumbi, earthquake level 3	**201%, fails**	**83%** with ground-floor shear walls

Every house carries its own weight on every slope and survives the cases outside its lesson. These numbers are regression tests. The model is educational, not a design code.

06 DESIGN PATTERNS

PATTERN	WHERE	WHY
State	**SimulationManager**, one **SimulationState** per state	Only legal input per state; pause keeps a shallow history

PATTERN	WHERE	WHY
Observer	**EventBus**, 24 typed event structs	UI, audio, autosave and the world label react without the simulation knowing them
Command and **Composite**	**PlaceComponentCommand** (a whole component), member remove, material change	Undo and redo of the last 20 actions; one undo removes a whole component
Strategy	**ILoadCase** ×5, **IScoringPolicy**	New load cases or scoring rules need no solver change
Factory Method	**MemberFactory**, **LoadCaseFactory**	One place maps a kind to its class and resolves slope geometry
Object Pool	**ObjectPool<T>** with **IPoolable**	No instantiate/destroy spikes during building, collapse and rain
Repository	**ISessionRepository**, **IProfileRepository**	Files and formats stay out of game logic
Flyweight	**MaterialData**, **HouseDesign** ScriptableObjects	Hundreds of members share one material and one design definition
Dependency injection	**AppBootstrap** composition root	Services are swapped for fakes in tests; no singletons

07 EVENT SYSTEM

EventBus keeps a list of handlers per message type; events are **readonly struct**s, so publishing allocates nothing. Publishers never hold references to listeners. A placement shows how one action fans out:

EVENT	PUBLISHED BY	HEARD BY
MemberPlaced	PlaceComponentCommand, once per piece	DataManager (placement log), AudioManager (one snap per component, haptic pulse), BuildHud
BuildProgressChanged, **StageCompleted**	TaskManager, once per component	BuildingState (next ghost), BuildHud, WorldHud (assembly %), autosave, audio, toasts
LoadTestCompleted	LoadTestManager	Results panel and stress chart, WorldHud (load stress), ScoreManager, DataManager, audio
StateChanged, **TrackingLostTooLong**	SimulationManager, ARManager	UIManager (screens), AudioManager (ambience), pause

EVENT	PUBLISHED BY	HEARD BY
StructureRevealed	SimulationManager, when the reveal sweep ends	AudioManager (reveal cue)

08 OBJECT POOL

ObjectPool<T> is generic over any component that implements **IPoolable** (**OnSpawned**, **OnDespawned**). It keeps inactive instances under a storage transform, can prewarm, and caps how many it keeps (**maxSize**), destroying the surplus. Every pool reports active, inactive and created counts to the debug overlay. **MemberViewFactory** keeps one pool per member shape (pad, strut, bar, panel, block, arch, shell) for built members and a separate pool per shape for ghosts, so a ghost is never mistaken for a member. Views reset their pivot, scale, colour and generated mesh on release; arch and shell views keep their mesh object and rebuild it in place. The effects controller pools debris (four pieces per failed member), rain drops and placement sparks.

The pools matter most in two places. The whole-house blueprint is re-drawn as faint hologram ghosts every time the selected component changes, which is up to 250 views for the flats. A collapse throws hundreds of debris pieces in under a second.

09 DATA PERSISTENCE

All data stays on the device, in line with the DMP. Learners only enter a display name.

FILE	HOLDS
profiles.json	Learner ids and display names
sessions/{learner}/{session}.json	One record per finished session: design, site, mode, every placement (accepted or rejected, with reason), every load test with peak utilisation and failures, score
autosave.json	The build in progress, after every stage and test
PlayerPrefs	Mute, last learner, debug overlay

- Writes go to a temporary file that is swapped in atomically (with a retry and a copy fallback on Windows), so a crash cannot leave half a file.
- Each record carries a schema version; newer versions are rejected, older ones can be migrated.
- The history screen lists past sessions, newest first, with score and result. The latest 50 sessions per learner are kept (more than the five required), and anything older than 180 days is purged.

- After a crash the next launch offers to resume the unfinished build. Reports export as JSON or CSV through the Android share sheet. Deleting a profile deletes its sessions.

10 AUDIO ARCHITECTURE

AudioManager listens to events and never to the simulation directly. It has three sources: effects for one-shot cues, ambience for the looping load-test sound, whose volume follows the slider, and world for birdsong while outdoors (placement, building, review). Sixteen cues (tap, snap, reject, stage and build complete, crack, collapse, survived, warning, pass and fail, quake and crowd loops) are synthesised by the editor's **AudioSynth**, so they carry no licence issue. Wind, rain and birds are recorded loops from a nature pack. Collapse waves play a crack and then falls, timed to the animation, with short haptic pulses. Mute lives in settings and also silences haptics. Clips load once through Addressables at start.

11 ASSET MANAGEMENT (ADDRESSABLES)

GROUP	CONTENTS	LOADED	RELEASED
Catalogue	Six designs (label **design**), twelve materials (label **material**)	App start	App exit
Audio	Cues and ambience loops (label **audio**)	App start	App exit
Interface	Rajdhani fonts	App start	App exit
Session Prefabs	Member prefabs, site plot with scenery, ghost and hologram materials, effects	Session start	Session end, restart, menu
Fit-outs	One bundle per house: furniture, lamps, the car's wheels and lights, the animals' eyes and ears	House revealed	Session end

ContentCatalogue records every handle under a scope (App, Session, Skin) and releases a whole scope at once. A revealed house loads only its own fit-out bundle. The PlayMode session test asserts that session and fit-out handle counts return to zero when the learner goes back to the menu, and the debug overlay shows live handle counts and pool use.



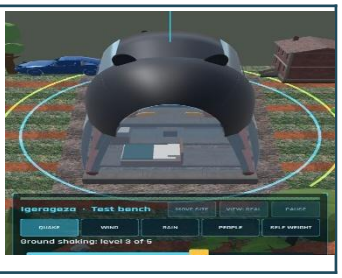
Building: the hologram



Reveal: turning real



Results: stress colours



Ingwe: rooms inside

12 DEVIATIONS FROM THE SRS

SRS SAID	BUILT	REASON
Unity 2022 LTS, AR Foundation 5	Unity 6000.4.7f1, AR Foundation 6.3, URP 17	The installed toolchain; current ARCore support and the custom URP hologram shader. No requirement changed.
Three designs: HX-01 Umutiba, HP-02 Umuheto, RF-04 Isoko	Six designs: three biomimetic, three Rwandan classics	Based on my reference boards. Each original lesson is kept: seismic bracing (car, panther, flats), wind uplift (elephant ears, bungalow roof) and heavy-rain drainage (elephant trunk).
Members placed one at a time	One card builds a whole component; members still snap to a ghost (FR-10) and follow stage and prerequisite rules (FR-09, FR-11)	Up to 257 members per house made one-at-a-time building slow. The solver and the session record still see every member.
Members drawn in their material	Built as a hologram; revealed as real materials when the test bench opens	Separates designing from testing and matches the reference theme. Stress colours show on the real materials.
Portrait and landscape	Portrait only	The tray and test bench need the height; one layout to test.
English and Kinyarwanda	English, with Kinyarwanda names and headings	Time. All strings are built in one UI layer, ready for localisation.
Import custom scenarios	Not built	Deferred; designs are already data (ScriptableObjects), so import is a loader, not a redesign.

13 TESTING AND BUILD

39 EditMode tests cover solver benchmarks against hand calculations (within 15%, NFR-14), solver behaviour (no mutation, collapse waves, bracing, slope), every lesson in the table in section 05, build-sequence and component rules, shell geometry and scoring. PlayMode tests drive a full car session in the real scene: setup, placement, a rejected out-of-order component, component building by drag and snap, undo and redo, a failed earthquake, reinforcement, a passing re-test, saving and asset release. A screenshot tour renders every house built and as a hologram. The Android build is IL2CPP, ARM64, OpenGL ES 3, minimum API 29, with ARCore required.

14 LIMITATIONS AND FUTURE IMPROVEMENTS

Limitations

- The solver is a simplified load-path model with calibrated factors, not finite-element analysis. Loads split evenly between supports, and stiffness sharing is approximate.
- Shell patches are checked as panels; true shell action (membrane forces, buckling of thin shells) is not modelled.
- Furniture is decorative and carries no load beyond the floor finish.
- Hologram transparency and glass sorting can flicker where many surfaces overlap.

Future improvements

- Cut openings in the shell for doors, and show crack patterns on walls instead of whole-member failure.
- Add Kinyarwanda, landscape layout and scenario import from the SRS.
- Let teachers author designs in an in-app editor, and share results with a class.
- Profile on low-end phones and add LODs to the larger houses.



Ingwe | **Inzovu** | **Imodoka** | **Urugo** | **Igorofa** | **Amacumbi**, hologram

Credits

Inspiration boards: Pinterest pins collected by the author (biomimetic and parametric architecture, holographic interface concepts). Models: 1994 Nissan 180MX by Grzybek (Poly Pizza, CC-BY). Textures: Bamboo 001B (ambientCG, CC0); brick, roof-tile, door and window textures from the Brick House pack; ground textures (GloverGames); metal plates, stylised wood and concrete packs; nature props and sounds from Unity Asset Store packs, which are not redistributed in the repository (listed in CREDITS.md). Font: Rajdhani (Indian Type Foundry, SIL OFL 1.1). Rwandan Flag and future patterns from google flow labs, the thousand-hills artwork and all sound effects.